Taylor and Maclaurin Series

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.02

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The Taylor series formula**

State the general Taylor series formula for $f(x)$ centered at $x = a$.

$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$.

2 **[Foundational]** • **Target: The Taylor series formula**

Find the Taylor series for $f(x) = \ln x$ centered at $a = 1$, up through the $(x-1)^3$ term.

$f(1) = 0$; $f'(x) = \frac{1}{x}$, $f'(1) = 1$; $f''(x) = -\frac{1}{x^2}$, $f''(1) = -1$; $f'''(x) = \frac{2}{x^3}$, $f'''(1) = 2$. Series: $0 + (x-1) - \frac{1}{2}(x-1)^2 + \frac{2}{6}(x-1)^3 = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}$.

3 **[Foundational]** • **Target: Common Maclaurin series**

State the Maclaurin series for e^x, $\sin x$, and $\cos x$.

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$. $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$.
$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$.

4 **[Foundational]** • **Target: Common Maclaurin series**

Use the Maclaurin series for e^x to find the Maclaurin series for e^{-x^2}.

Substitute $x \to -x^2$: $e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!} = 1 - x^2 + \frac{x^4}{2} - \cdots$.

5 **[Foundational]** • **Target: Using known series to build new series**

Use the Maclaurin series for $\sin x$ to find the first three nonzero terms of the Maclaurin series for $x\sin x$.

$x\sin x = x\left(x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots\right) = x^2 - \frac{x^4}{6} + \frac{x^6}{120} - \cdots$.

6 **[Foundational]** • **Target: Using known series to build new series**

Use the known Taylor series for $\cos x$ to estimate $\cos(0.1)$ using the first three nonzero terms.

$\cos(0.1) \approx 1 - \frac{(0.1)^2}{2} + \frac{(0.1)^4}{24} = 1 - 0.005 + 0.0000042 \approx 0.995004$.

7 **[Foundational]** • **Target: The Lagrange error bound**

State the Lagrange Error Bound formula for the remainder $R_n(x)$ of a Taylor polynomial of degree n.

$|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$, where M is an upper bound for $|f^{(n+1)}(z)|$ on the interval between a and x.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: The Lagrange error bound**

Use the Lagrange Error Bound to estimate the maximum error when approximating $\sin(0.5)$ with the Taylor polynomial $P_3(x) = x - \frac{x^3}{6}$ (degree 3, centered at 0), given $|f^{(4)}(z)| \le 1$ for all z.

$|R_3(0.5)| \le \frac{1}{4!}|0.5|^4 = \frac{0.0625}{24} \approx 0.0026$.

9 **[Intermediate]** • **Target: Radius of convergence of Taylor series**

State the radius of convergence of the Maclaurin series for e^x, $\sin x$, and $\cos x$.

All three series have radius of convergence $R = \infty$ — they converge for all real x.

10 **[Intermediate]** • **Target: Visualizing Taylor polynomial approximations**

Compare $f(x) = \sin x$ to its degree-5 Maclaurin polynomial $P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$ on $[-\pi, \pi]$, and describe where the approximation is best and worst.

The approximation is excellent near $x = 0$ (where the curves are nearly indistinguishable) and progressively worsens toward $x = \pm\pi$, since Taylor polynomials are most accurate near the center of expansion and lose accuracy farther away.

11 **[Intermediate]** • **Target: Visualizing Taylor polynomial approximations**

Compare $f(x) = e^x$ to its degree-2 Maclaurin polynomial $P_2(x) = 1 + x + \frac{x^2}{2}$ on $[-2, 2]$, and identify visually where P_2 underestimates or overestimates e^x.

Since e^x is convex (concave up) and its Taylor polynomial is tangent to it at $x = 0$ up to second order, $P_2(x)$ tends to underestimate e^x away from the center, with the gap growing faster on the positive side due to the exponential's rapid growth.

12 **[Intermediate]** • **Target: Building Taylor series at a nonzero center**

Find the Taylor series for $f(x) = \cos x$ centered at $a = \frac{\pi}{2}$, up through the cubic term.

$f(\frac{\pi}{2}) = 0$; $f'(x) = -\sin x$, $f'(\frac{\pi}{2}) = -1$; $f''(x) = -\cos x$, $f''(\frac{\pi}{2}) = 0$; $f'''(x) = \sin x$, $f'''(\frac{\pi}{2}) = 1$. Series: $-(x - \frac{\pi}{2}) + \frac{1}{6}(x - \frac{\pi}{2})^3 - \cdots$.

13 **[Intermediate]** • **Target: Applying the Lagrange Error Bound to bound accuracy requirements**

Using the degree-4 Maclaurin polynomial for $\cos x$ and the bound $|f^{(5)}(z)| \le 1$, find the maximum error in approximating $\cos(0.3)$.

$|R_4(0.3)| \le \frac{1}{5!}|0.3|^5 = \frac{0.00243}{120} \approx 0.0000203$ — an extremely small error, confirming the Maclaurin polynomial is highly accurate this close to the center.

© ClipSAT — clipsat.org | All Rights Reserved

14 **[Intermediate]** • **Target: The binomial series**

State the binomial series for $(1+x)^k$ up through the quadratic term, and use it to find the Maclaurin series for $\sqrt{1+x}$ (i.e. $k=\frac{1}{2}$) up through the quadratic term.

$(1+x)^k = 1 + kx + \frac{k(k-1)}{2}x^2 + \cdots$. With $k=\frac{1}{2}$: $\sqrt{1+x} = 1 + \frac{1}{2}x + \frac{(1/2)(-1/2)}{2}x^2 + \cdots = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: The binomial series** • **Cross-Lesson: 3.02 Vector-Valued Functions**

Use the series found above to approximate $\sqrt{1.1}$ using the first three terms.

$\sqrt{1.1} \approx 1 + \frac{0.1}{2} - \frac{(0.1)^2}{8} = 1 + 0.05 - 0.00125 = 1.04875$ (true value ≈ 1.04881).

16 **[Challenging]** • **Target: Deriving new Maclaurin series from known ones** • **Cross-Lesson: 4.01 Polar Curves and Derivatives**

Use the known series for e^x to find the Maclaurin series for $\cosh x = \frac{e^x + e^{-x}}{2}$, up through the quartic term.

$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots$; $e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} - \cdots$. Averaging cancels odd terms: $\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{24} + \cdots$.

17 **[Challenging]** • **Target: Building Taylor series at a nonzero center** • **Cross-Lesson: 4.02 Area in Polar Coordinates**

Level 3 · Advanced Let $f(x)$ be a function with $f(2)=3$, $f'(2)=-1$, $f''(2)=4$, and $f'''(2)=-12$. Write the third-degree Taylor polynomial $P_3(x)$ for f centered at $x=2$, and use it to approximate $f(2.1)$.

Construct the Taylor polynomial using $a=2$: $P_3(x) = f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3$ $= 3 - 1(x-2) + \frac{4}{2}(x-2)^2 + \frac{-12}{6}(x-2)^3 = 3 - (x-2) + 2(x-2)^2 - 2(x-2)^3$ Substitute $x=2.1$, so $x-2=0.1$: $P_3(2.1) = 3 - 0.1 + 2(0.1)^2 - 2(0.1)^3 = 3 - 0.1 + 2(0.01) - 2(0.001)$ $= 3 - 0.1 + 0.02 - 0.002 = 2.918$

18 **[Challenging]** • **Target: Applying the Lagrange Error Bound to bound accuracy requirements** • **Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests**

Level 2 · Medium Find the third-degree Maclaurin polynomial $P_3(x)$ for $f(x)=\cos x$.

Compute derivatives at $x=0$: • $f(0)=\cos(0)=1$ • $f'(0)=-\sin(0)=0$ • $f''(0)=-\cos(0)=-1$ • $f'''(0)=\sin(0)=0$ Applying the Maclaurin formula: $P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 = 1 - \frac{x^2}{2}$. (Notice the cubic term has coefficient 0).

© ClipSAT — clipsat.org | All Rights Reserved

19 [Challenging] • Target: The binomial series • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

Set up Lagrange system $ablaf = \lambda ablag$: $ablaf = \langle 2x, 4y \rangle$, $ablag = \langle 2x, 2y \rangle$. 1) $2x = 2\lambda x \Rightarrow 2x(1 - \lambda) = 0$ 2) $4y = 2\lambda y \Rightarrow 2y(2 - \lambda) = 0$ 3) $x^2 + y^2 = 1$ From (1), either $x = 0$ or $\lambda = 1$: • If $x = 0$: from constraint (3), $y^2 = 1 \Rightarrow y = \pm 1$. Points: $(0, 1)$ and $(0, -1)$. Value: $f(0, \pm 1) = 0 + 2(1)^2 = 2$. • If $\lambda = 1$: from (2), $2y(2 - 1) = 2y = 0 \Rightarrow y = 0$. From constraint (3).

20 [Challenging] • Target: Deriving new Maclaurin series from known ones • Cross-Lesson: 6.01 Power Series and Interval of Convergence

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n \to \infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

(a) Lower bound via AM-GM: For any positive real number $x_n > 0$, the Arithmetic Mean-Geometric Mean inequality states that $\frac{a+b}{2} \geq \sqrt{ab}$. With $a = x_n$ and $b = \frac{5}{x_n}$: $x_{n+1} = \frac{x_n + 5/x_n}{2} \geq \sqrt{x_n \cdot \frac{5}{x_n}} = \sqrt{5}$. Thus $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Monotonicity & Convergence: Compute the difference: $x_{n+1} - x_n = \frac{1}{2}x_n + \frac{5}{2x_n} - x_n = \frac{5 - x_n^2}{2x_n}$. Since $x_n \geq \sqrt{5}$ for $n \geq 1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Taylor series formula

Use the Maclaurin series for e^u (with $u = 3x^2$) to find the coefficient of x^8 in the Maclaurin series for e^{3x^2}.

a. $\frac{2187}{8}$

b. $\frac{9}{4480}$

c. $\frac{35}{8}$

d. $\frac{27}{8}$ ✓

Substituting $u = 3x^2$ into $e^u = \sum \frac{u^n}{n!}$: the x^8 term comes from $n = 4$: $\frac{(3x^2)^4}{4!} = \frac{81}{24}x^8$. Coefficient: $\frac{27}{8}$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Common Maclaurin series

Using the Lagrange Error Bound $|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M=6$, $n=2$, and $|x-a|=1$, find the error bound.

a. 2

b. $\frac{7}{6}$

c. 3

d. 1 ✓

$|R_2(x)| \le \frac{6}{(2+1)!}(1)^{2+1} = \frac{6}{6} = 1$.

23 [Foundational] • Target: Using known series to build new series

The Maclaurin series for $\sqrt{1+x}$ is a special case of the binomial series with $\alpha =$

a. 2

b. 1

c. $-1/2$

d. $1/2$ ✓

Applying Convergence Tests and Power Series principles, $(1+x)^{1/2}$. Therefore, the verified correct option is $1/2$.

24 [Foundational] • Target: The Lagrange error bound

Taylor polynomial $T_2(x)$ for $f(x) = \cos x$ at $x = 0$:

a. $1 - x$

b. $1 - x^2$

c. $1 - x^2/2$ ✓

d. $x - x^3/6$

$\cos x \approx 1 - x^2/2$ (second-order).

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Radius of convergence of Taylor series

The rate of change of the volume of a spherical balloon being inflated is given by $\frac{dV}{dt} = 12\pi$ cm^3/s. If the initial volume of the balloon is 36π cm^3, find the volume of the balloon after t seconds. The volume V of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.

a. $V(t) = 12\pi t + 36\pi$ ✓

b. $V(t) = \frac{4}{3}\pi t^3 + 36\pi$

c. $V(t) = 4\pi t^3 + 36\pi$

d. $V(t) = 12\pi t^3 + 36\pi$

We are given that $\frac{dV}{dt} = 12\pi$. To find the volume $V(t)$, we need to integrate $\frac{dV}{dt}$ with respect to t. $V(t) = \int \frac{dV}{dt} dt = \int 12\pi dt$ $V(t) = 12\pi t + C$ We are given that the initial volume is 36π cm^3, which means $V(0) = 36\pi$. Substituting $t = 0$ into the equation for $V(t)$: $36\pi = 12\pi(0) + C$ $C = 36\pi$ Therefore, the volume of the balloon after t seconds is $V(t) = 12\pi t + 36\pi$.

26 [Foundational] • Target: Visualizing Taylor polynomial approximations

Taylor series of $1/(1 - x)$ at $x = 0$:

a. $\sum x^n/n!$

b. $\sum (-1)^n x^n$

c. $\sum x^{n+1}$

d. $\sum x^n$ **for** $|x| < 1$ ✓

Applying Convergence Tests and Power Series principles, Geometric series. Therefore, the verified correct option is $\sum x^n$ for $|x| < 1$.

27 [Foundational] • Target: Building Taylor series at a nonzero center

Radius of convergence of $\sum n!\, x^n$:

a. ∞

b. 1

c. 0 ✓

d. e

Applying Sequences and Series principles, $R = 1/\limsup |n!\,|^{1/n} \to 0$. Therefore, the verified correct option is 0.

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Applying the Lagrange Error Bound to bound accuracy requirements

Using the first three terms of the Maclaurin series for e^x, approximate $e^{1 \times \frac{1}{4}}$ as a fraction.

a. $\frac{173}{160}$

b. $\frac{41}{32}$ ✓

c. $\frac{253}{160}$

d. $\frac{237}{160}$

$e^x \approx 1 + x + \frac{(x)^2}{2}$, at $x = \frac{1}{4}$: $1 + \frac{1}{4} + \frac{\frac{1}{16}}{2} = \frac{41}{32}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The binomial series

Use the degree-3 Maclaurin polynomial of e^x to approximate its value at $x = \frac{1}{10}$.

a. $\frac{221}{200}$

b. $\frac{6631}{6000}$ ✓

c. $\frac{6637}{6000}$

d. $\frac{6031}{6000}$

$T_3(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$; at $x = \frac{1}{10}$ this gives $\frac{6631}{6000}$.

30 [Intermediate] • Target: Deriving new Maclaurin series from known ones

The Maclaurin series for e^{-x^2} (substitute $-x^2$ for x in e^x series):

a. $\sum \frac{(-1)^n x^{2n}}{n!}$ ✓

b. $\sum x^{2n}/n!$

c. $\sum (-x)^n/n!$

d. $\sum x^n/n!$

$e^{-x^2} = \sum \frac{(-x^2)^n}{n!} = \sum \frac{(-1)^n x^{2n}}{n!}$.

31 [Intermediate] • Target: The Taylor series formula

Sum $\sum_{n=1}^{\infty} n(n-1)x^n/(n!)$ simplifies using e^x series to:

a. $e^x(x^2 - x)$

b. xe^x

c. x^2e^x ✓

d. $x^2e^x/2$

$n(n-1)/n! = 1/(n-2)! \Rightarrow x^2 \sum x^{n-2}/(n-2)! = x^2e^x$.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Common Maclaurin series

The power series $\sum a_n x^n$ represents a function that is C^∞ (infinitely differentiable) inside its radius of convergence. This means all derivatives:

a. are polynomials

b. are zero

c. **exist ✓**

d. diverge

Power series define smooth (C^∞) functions inside R.

33 [Intermediate] • Target: Using known series to build new series

Taylor polynomial error: $|f(x) - T_n(x)| \leq \frac{M|x-a|^{n+1}}{(n+1)!}$ where $M =$

a. $\max|f^{(n+1)}|$ **on interval ✓**

b. $|f^{(n)}(a)|$

c. $n!$

d. $|f(a)|$

Lagrange error bound uses max of $(n+1)$th derivative.

34 [Intermediate] • Target: The Lagrange error bound

Taylor series of $f(x) = (1+x)^\alpha$ (binomial series) for $|x| < 1$:

a. $1 + \alpha x$ only

b. $\sum e^{\alpha x}$

c. $\sum x^n / n$

d. $\sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$ ✓

Binomial series: $\sum \binom{\alpha}{n} x^n$, $|x| < 1$.

35 [Intermediate] • Target: Radius of convergence of Taylor series

Power series $\sum \frac{(x-2)^n}{3^n}$: centre and radius:

a. centre 0, $R = 3$

b. centre 3, $R = 2$

c. centre 2, $R = 1/3$

d. **centre 2,** $R = 3$ ✓

Centre $x = 2$, ratio $r = |(x-2)/3| < 1 \Rightarrow R = 3$.

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Visualizing Taylor polynomial approximations

Using power series, compute $\lim_{x\to 0}\frac{1-\cos x}{x^2}$:

a. 1/2 ✓

b. 1

c. 0

d. 2

$1-\cos x \approx x^2/2 \Rightarrow (x^2/2)/x^2 = 1/2$.

37 [Intermediate] • Target: Building Taylor series at a nonzero center

Power series $\sum_{n=0}^{\infty} n(x-1)^n$. Radius of convergence:

a. 0

b. 1 ✓

c. 2

d. ∞

$L = \lim n^{1/n}\cdot|x-1| \to |x-1|; R = 1$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applying the Lagrange Error Bound to bound accuracy requirements • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Taylor remainder (Lagrange form): $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$ for some c between a and x. This bounds the:

a. average error

b. maximum error

c. minimum error

d. **exact error** ✓

Lagrange form gives exact remainder at some c.

39 [Challenging] • Target: The binomial series • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Hypergeometric series ${}_2F_1(a,b;c;x) = \sum_{n=0}^{\infty}\frac{(a)_n(b)_n}{(c)_n n!}x^n$ (Pochhammer symbol). Radius of convergence:

a. depends on a, b, c

b. always ∞

c. **always 1** ✓

d. always 0

Gauss hypergeometric: radius of convergence 1.

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Deriving new Maclaurin series from known ones • Cross-Lesson: 6.01 Power Series and Interval of Convergence

If a function equals its Taylor series everywhere, it is called:

a. **analytic** ✓

b. polynomial

c. differentiable

d. continuous

Analytic means equal to its power series.

41 [Challenging] • Target: The Taylor series formula • Cross-Lesson: 1.01 Integration by Parts

Maclaurin series for e^x: first four terms:

a. $1 + x + x^2 + x^3$

b. $1 + x + x^2/2! + x^3/3!$ ✓

c. $x + x^2 + x^3$

d. $1 - x + x^2 - x^3$

$e^x = \sum x^n/n! = 1 + x + x^2/2 + x^3/6 + \cdots$

42 [Challenging] • Target: Common Maclaurin series • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

$\lim_{x \to 0} \frac{1 - \cos x}{x^2}$. Using L'Hôpital twice or Taylor series:

a. $1/2$ ✓

b. 0

c. 1

d. $-1/2$

$1 - \cos x \approx x^2/2 \Rightarrow 1/2$.

43 [Challenging] • Target: Using known series to build new series • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Bessel function $J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{x}{2}\right)^{2m}$. Radius of convergence:

a. 1

b. ∞ ✓

c. π

d. 2

Ratio test: $L = 0 \Rightarrow R = \infty$.

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: The Lagrange error bound • Cross-Lesson: 3.01 Parametric Equations

The Padé approximant $[m/n]$ for $f(x)$ is a rational function $P_m(x)/Q_n(x)$ matching the Taylor series to order $m+n$. It typically converges more:

a. slowly than Taylor

b. only at 0

c. **rapidly than Taylor for poles** ✓

d. never

Padé: better than Taylor near poles.

45 [Challenging] • Target: Radius of convergence of Taylor series • Cross-Lesson: 3.02 Vector-Valued Functions

The binomial series $(1+x)^{1/2}$ up to x^2:

a. $1+x-x^2$

b. $1+x/2+x^2/8$

c. $1+x/2$

d. $1+x/2-x^2/8$ ✓

$(1/2)(1/2-1)/2! \cdot x^2 = -x^2/8$.

© ClipSAT — clipsat.org | All Rights Reserved